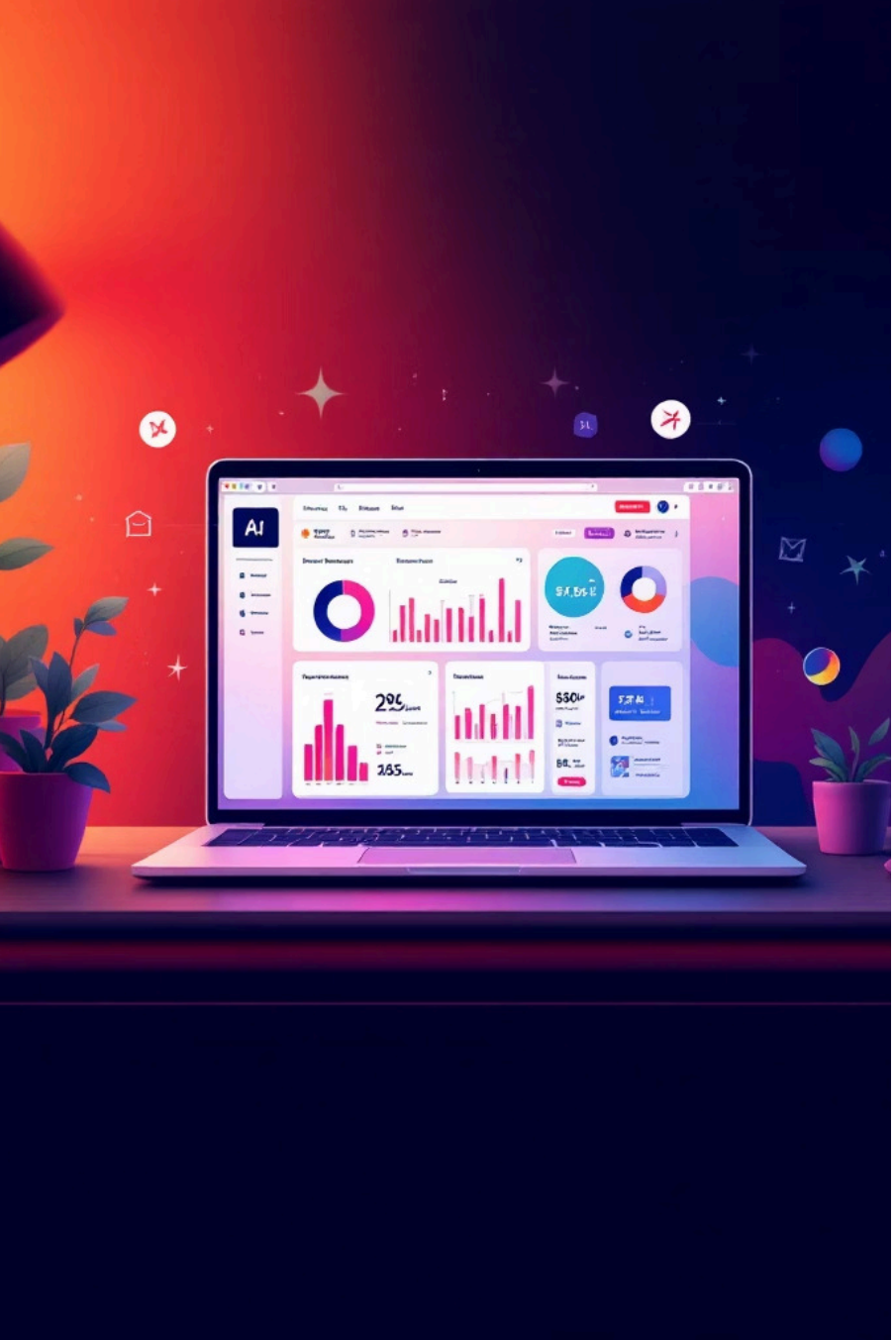

Building AI Applications with Streamlit

Phase 2 – GenAI Frameworks & UI Development

Learn how to transform your Python AI scripts into interactive web applications that anyone can use. Streamlit makes it easy to build beautiful, functional interfaces without web development expertise.

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What is Streamlit?

Streamlit is a powerful Python framework designed specifically for building web applications for AI, data science, and machine learning projects.

Key Benefits

- No HTML or CSS required
- Rapid prototyping and iteration
- Seamless integration with AI models
- Automatic page reloading

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Why Streamlit for AI Projects

Streamlit has become the go-to tool for AI developers who want to create user-friendly interfaces for their machine learning models and generative AI applications.

AI Chatbots

Build conversational interfaces that connect to LLMs like GPT-4

Content Generators

Create AI-powered blog generators, resume builders, and writing assistants

Data Summarizers

Develop tools that analyze and summarize documents, articles, and research papers

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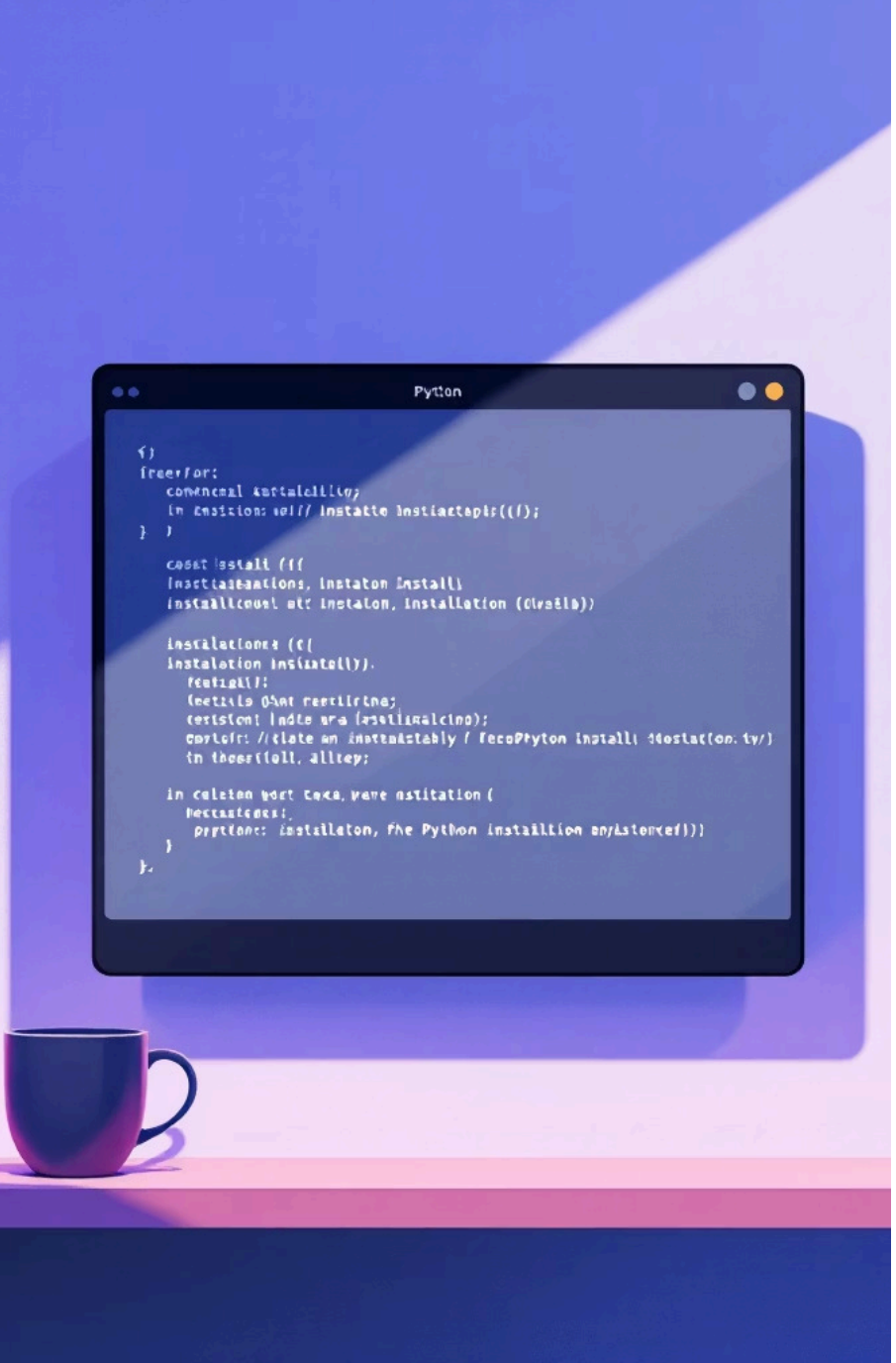

Installing Streamlit

Getting started with Streamlit is incredibly simple. You'll need Python installed on your system first, then follow these straightforward steps.

Install Streamlit

Open your terminal and run:

```
pip install streamlit
```

Verify Installation

Test that Streamlit works:

```
streamlit hello
```

This launches a demo web app in your browser

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Running a Streamlit App

Creating Your App

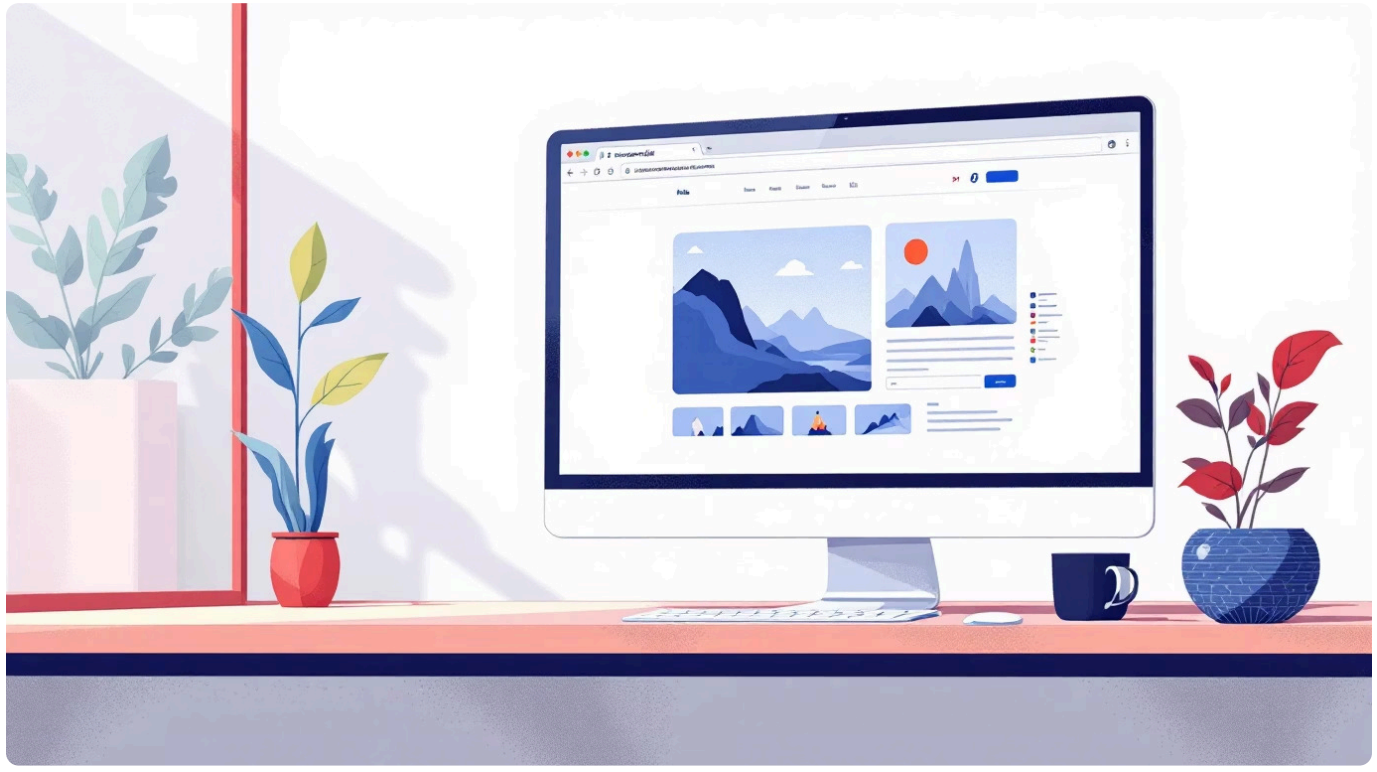
Create a Python file called `app.py` in your project folder. This file will contain all your Streamlit code.

Launching the App

Run this command in your terminal:

```
streamlit run app.py
```

Streamlit starts a local web server and opens your app in the default browser at `http://localhost:8501`



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Basic Streamlit Code Example

Let's create a simple interactive application that greets users by name. This example demonstrates the core concepts of Streamlit development.

```
import streamlit as st

st.title("My First AI App")

name = st.text_input("Enter your name")

if name:
    st.write("Hello", name, "!")
```

st.title()

Creates the main page heading displayed at the top of your application

st.text_input()

Generates a text box where users can enter information

st.write()

Displays text, variables, or other content on the page

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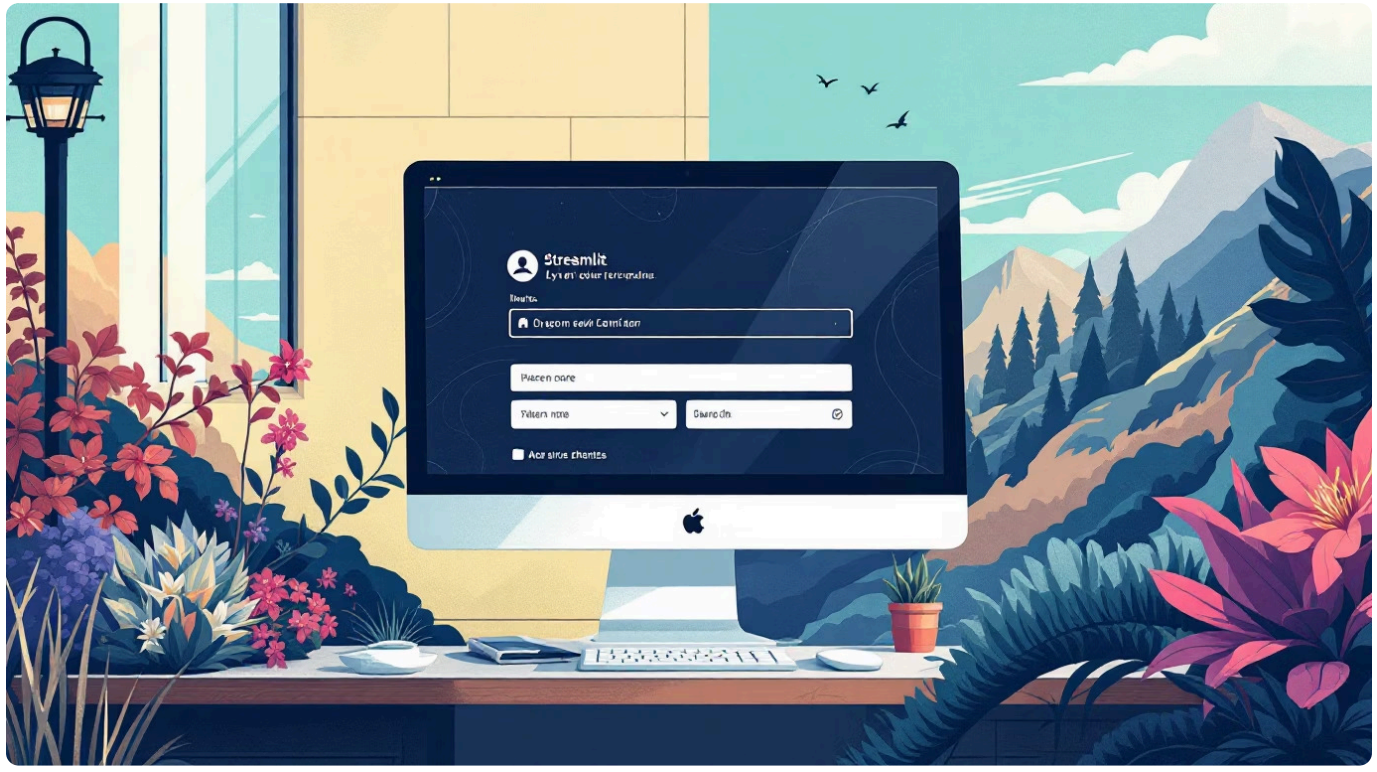
Example Output of the App

What You'll See

When you run the code, Streamlit opens a web page with:

- A title heading "My First AI App"
- A text input field labeled "Enter your name"
- Dynamic greeting that appears as you type

Streamlit automatically updates the page whenever input changes, creating a responsive experience without additional code.



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Streamlit UI Components

Streamlit provides a rich set of interactive components to create engaging user interfaces for your AI applications.

st.text_input()

Text box for user input with optional default values and validation

st.button()

Action button that triggers specific functionality when clicked

st.selectbox()

Dropdown menu for selecting from predefined options

st.slider()

Interactive slider for selecting numeric values within a range

st.write()

Flexible display method for showing text, variables, and results

st.file_uploader()

File upload component for processing documents and images

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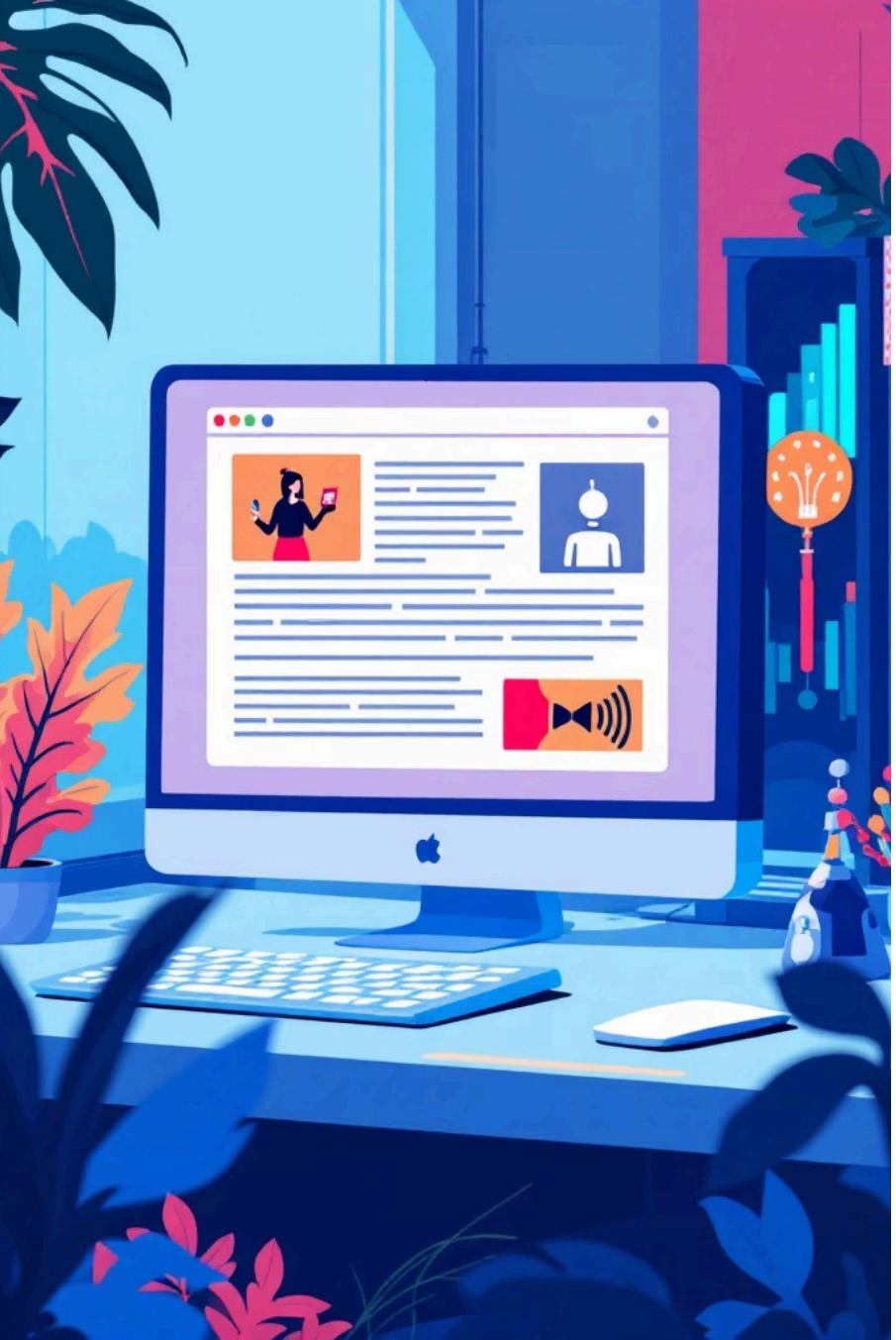

Example AI Application

AI Blog Generator

This simplified example shows how to create a foundation for an AI-powered content generator. Later, you can connect it to actual AI APIs like OpenAI's GPT to generate real blog posts.

```
import streamlit as st

st.title("AI Blog Generator")

topic = st.text_input("Enter Blog Topic")

if st.button("Generate"):
 st.write("Generating blog about", topic)
 # In a complete version, call an AI API here
 # to generate actual blog content based on the topic
```

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Capstone Project

Now it's your turn to build a complete AI web application using Streamlit. This capstone project will showcase everything you've learned in this course.

AI Chatbot

Create a conversational interface that connects to large language models and responds to user queries

AI Blog Generator

Build a tool that generates complete blog posts from topic ideas using AI language models

AI Resume Generator

Develop an application that creates professional resumes from user-provided information

AI Study Assistant

Design a study tool that summarizes notes, generates flashcards, and creates practice quizzes

Final Goal: Present a fully working AI web application built from scratch that demonstrates your understanding of Streamlit, AI integration, and user interface design principles.

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